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STUDENTS ASSOCIATION

MUSA (MUMBAI UNIVERSITY STUDENTS ASSOCIATION)

QUESTION BANK OF COMPUTER ORGANIZATION & ARCHITECTURE

S.E SEM-III **SCHEME: NEP SCHEME** **FOR WINTER SESSION 2025**

BRANCH: COMPS/AIML/AIDS/IOT/CS/DS

Q.1) Explain Flynn's classification in detail.

Q.2) Explain various pipeline hazards with example.

Q.3) List and explain various characteristics of memory.

Q.4) Explain different addressing modes of 8086 with examples.

Q.5) Discuss various cache memory mapping techniques with diagram and state the advantages and disadvantages of it.

Q.6) Explain the instruction cycle with the help of a neat state diagram.

Q.6) With a neat diagram, explain the six stage instruction pipeline

Q.7) Explain Hardwired control unit. List and explain various methods to design hardwired control unit.

Q.8) What is bus arbitration? Describe various Bus arbitration methods.

Q.9) Explain the architecture of 8086 with neat diagram.

Q.10) Explain design of control unit w.r.t. microprogrammed and hardwired approach.

Q.11) Describe DMA and explain its various data transfer techniques.

Q.12) Describe in detail Von-Neuman model with a neat block diagram.

Q.13) Explain the IEEE 754 standards for representation of floating point numbers.

Q.14) Explain Register organization of processor.

Q.15) Explain the concept of Cache coherence and techniques to resolve it.

Q.16) Describe Interleaved and Associative memory. ***(Note: This question may ask in difference between form also).***

Q.17) Draw and explain microinstruction sequencing and execution organization.

Q.18) Explain buses in I/O organization and explain different types of Buses.

Q.19) Explain virtual memory in detail. Describe paging and segmentation with neat diagrams.

Q.20) List and explain different types of ROM with their characteristics.

Q.21) Explain the concept of an I/O Interface. Describe the functions of I/O modules with a neat block diagram.

Q.22) Draw and explain the instruction formats of 8086 with suitable examples.

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Q.19) **Differentiate between the following:**

i) Hardwired control unit and Microprogrammed control unit
ii) SRAM and DRAM
iii) RISC and CISC
iv) Real mode, protected mode and virtual mode ***(Prepare for explanation part also).***
v) Programmed I/O, Interrupt-driven I/O

Q.20) Write a short note on:

i) Mitigation of hazards with branch prediction and data forwarding techniques.
ii) Amdahl's laws
iii) Logic gates
iv) Draw and explain memory hierarchy.

Numerical

Q.1) Draw the flowcharts of booths algorithm for signed integer multiplication and perform the multiplication between the numbers using booths algorithm:

i) -6 and 2 **OR** 6 and 2
ii) 6 and 2
iii) 5 and 2 **OR** -5 and -3
iv) 3 and -2
v) $(10)_{10}$ with $(8)_8$

Q.2) Draw the flowchart for Restoring and Non restoring division algorithm and perform. ***(Number will be given to you)***

Q.3) Convert the following number to IEEE 754 Standard of Single and Double precision format.

Q.4) Conversion of: Binary, octal, decimal, hexadecimal.

Q.5) Perform the binary numbers using 1's and 2's complement

*****ALL THE BEST*****